

MMR-FinMath: A Multimodal Benchmark and Fine-Grained Evaluation Framework for Financial Mathematical Reasoning

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Abstract

Joint visual-textual reasoning is fundamental to financial intelligence. However, existing benchmarks for Multimodal Large Reasoning Models (MLRMs) often focus on general domains or rely on shallow visual cues, leaving the rigorous assessment of complex financial reasoning largely under-explored. To bridge this gap, we introduce MMR-FinMath, a comprehensive benchmark designed to evaluate the complex mathematical reasoning abilities of MLRMs in financial scenarios. MMR-FinMath requires four challenging capabilities: expert-level visual perception, implicit knowledge retrieval, cross-modal long-chain reasoning, and high-precision numerical computation. Correspondingly, we further propose a fine-grained evaluation framework tailored to these four dimensions, enabling precise identification of reasoning bottlenecks in MLRMs. Extensive experiments show that even state-of-the-art MLRMs struggle on these challenging tasks, underscoring the need to advance cross-modal deep reasoning for trustworthy financial intelligence.

Overall Framework

The overall construction pipeline of MMR-FinMath is summarized in Figure 1.

MMR-FinMath is sourced from six categories of authoritative examinations, including CFA, FRM, and major Chinese-language exams. We curate problems that genuinely require advanced reasoning, and

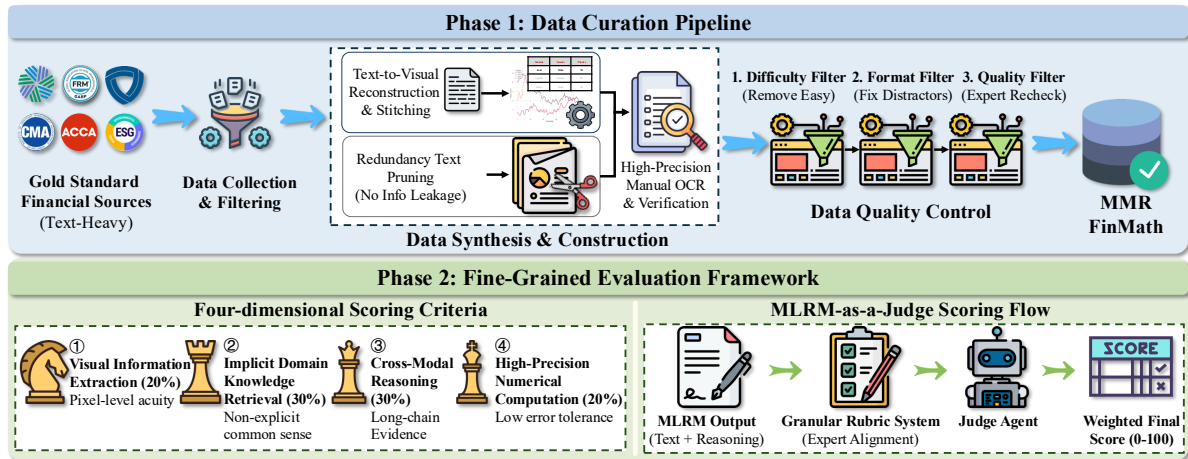


Figure 1: Overall Framework of MMR-FinMath.

construct 1,500 bilingual (Chinese–English) questions via numeric-to-figure rendering, image stitching, and manual OCR verification. The benchmark spans 11 subfields and 9 categories of complex figures. On the evaluation side, we decompose performance into four weighted dimensions: visual extraction, implicit domain knowledge, cross-modal chain-of-thought reasoning, and high-precision computation. Scoring is conducted by a judge model using 4,849 expert-aligned rubrics to pinpoint bottlenecks. A representative data instance in FinMathVL is provided in Figure 2.

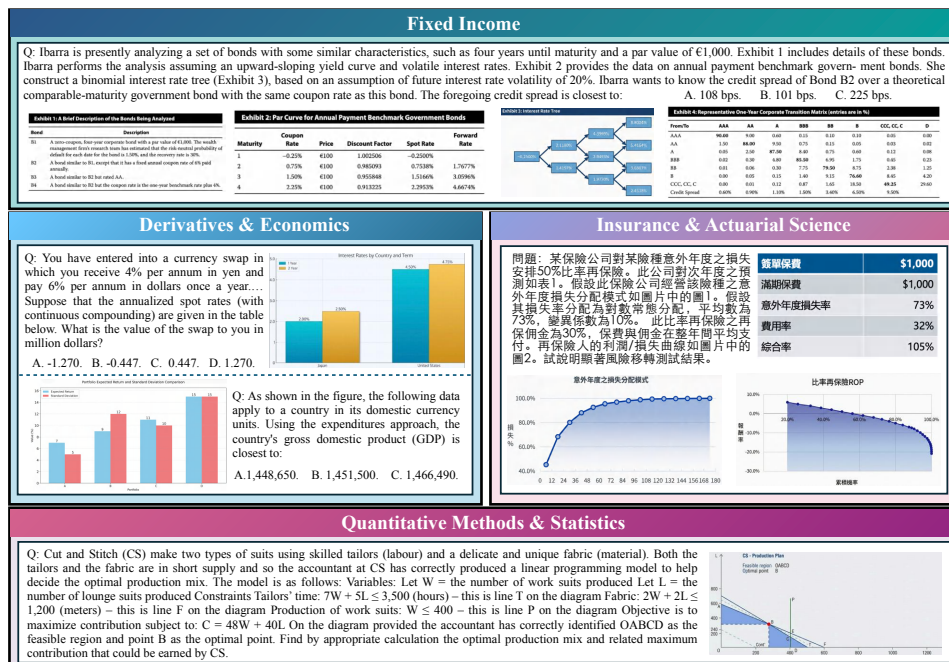


Figure 2: Representative examples from the MMR-FinMath benchmark across core financial sub-domains in two languages (i.e., English and Chinese). Each question is categorized by the financial mathematical skill it measures to enable detailed analysis.

Selected Experimental Results

Evaluation across 17 open-source and closed-source MLRMs shows that the benchmark remains highly challenging: even leading models achieve only about 57% overall accuracy on Multiple-Choice Questions and 64% on Question Answering, as shown in Table 1. The gap is particularly pronounced in actuarial science and insurance, where open-source models score consistently low and exhibit a large drop from multiple-choice performance to generative derivations.

Model	MCQ								QA				
	Derivs.	Econ.	Equities	FSA	FI	Portfolio	Quant	Overall	Acct.	Corp.	Actuarial	Quant	Overall
<i>Open-source MLRMs</i>													
VL-Rethinker-7B	37.11	45.71	31.69	41.18	31.72	30.83	39.53	35.50	38.34	47.58	16.99	39.71	26.62
Keye-VL-1.5-8B	37.11	48.57	26.76	42.02	35.07	30.08	44.19	35.25	36.65	45.75	28.14	38.62	32.78
MiMo-VL-7B-RL	49.48	45.71	38.03	58.82	50.75	47.37	51.94	51.67	66.66	66.42	28.21	74.58	44.53
Qwen3-VL-4B	23.20	22.86	24.65	35.29	23.13	38.35	38.76	29.00	63.00	71.92	27.65	66.42	43.18
Qwen3-VL-8B	50.00	45.71	44.37	55.46	43.66	51.88	65.12	51.67	67.45	74.83	31.69	64.71	46.54
Qwen3-VL-30B-A3B	61.26	31.43	<u>49.30</u>	60.50	<u>55.60</u>	56.39	67.44	58.78	70.97	77.12	<u>33.02</u>	<u>72.54</u>	49.05
Skywork-R1V-38B	41.24	45.71	42.96	38.66	41.42	51.13	43.41	43.83	48.42	51.50	19.78	56.88	32.51
VL-Rethinker-72B	51.03	42.86	38.03	57.14	45.15	53.38	49.61	48.33	57.32	63.21	29.13	52.25	40.84
QVQ-72B-Preview	46.91	62.86	38.73	52.94	44.78	43.61	53.49	48.17	60.45	69.83	33.16	56.96	44.88
GLM-4.6V	<u>55.96</u>	<u>60.00</u>	53.96	<u>59.66</u>	56.34	<u>54.89</u>	<u>64.34</u>	<u>58.36</u>	<u>69.77</u>	<u>76.17</u>	31.62	66.92	<u>47.39</u>
<i>Closed-source MLRMs</i>													
Qwen3-VL-Plus	64.43	60.00	47.18	67.23	60.45	59.40	61.24	62.33	73.92	77.00	39.16	72.71	53.51
Doubao-Seed-1.8	77.32	74.29	<u>65.49</u>	<u>73.11</u>	74.81	<u>64.12</u>	70.54	73.49	77.54	83.25	39.08	80.00	55.13
Claude Sonnet 4.5	64.95	65.71	54.23	53.78	57.09	48.12	58.14	58.08	79.35	<u>89.12</u>	44.52	<u>82.42</u>	<u>59.89</u>
GPT-4o	35.57	28.57	40.14	37.82	28.73	27.07	45.74	34.83	56.21	66.25	25.65	56.96	39.20
GPT-5.2	62.89	51.43	54.93	55.46	55.22	41.35	61.24	57.33	<u>79.86</u>	89.50	50.98	85.17	63.95
o3	<u>74.74</u>	68.57	61.27	71.43	67.16	71.43	<u>62.02</u>	69.92	74.76	80.88	44.15	66.04	56.44
Gemini 3 Flash	72.16	<u>71.43</u>	66.20	81.51	<u>72.01</u>	63.91	57.36	<u>72.17</u>	80.04	84.04	<u>45.02</u>	77.25	59.48

Table 1: Evaluation results of state-of-the-art MLRMs.

Ablation studies further show that text-only settings often outperform image-only, and even image-plus-text, indicating that visual numerical extraction and cross-modal alignment are the primary bottlenecks. Incorporating RAG-style retrieval of terminology and formulas yields stable gains. Error analysis finds that retrieval failures and reasoning errors account for the largest share of mistakes, highlighting clear directions for improvement.